Solutions: Linear Algebra

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Exercise 4.2

1. $A = \begin{bmatrix} 7 & -1 \\ 6 & 9 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix}$ $C = \begin{bmatrix} 8 & 3 \\ 6 & 1 \end{bmatrix}$

   (a) $A + B = \begin{bmatrix} 7 & 3 \\ 9 & 7 \end{bmatrix}$

   (b) $C - A = \begin{bmatrix} 1 & 4 \\ 0 & -8 \end{bmatrix}$

   (c) $3A = \begin{bmatrix} 21 & -3 \\ 18 & 27 \end{bmatrix}$

   (d) $4B + 2C = \begin{bmatrix} 0 & 16 \\ 12 & -8 \end{bmatrix} + \begin{bmatrix} 16 & 6 \\ 12 & 2 \end{bmatrix} = \begin{bmatrix} 16 & 22 \\ 24 & -6 \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 8 \\ 3 & 0 \\ 5 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}$ $C = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$

   (a) AB is defined as number of columns in A is two which is equal to the number of rows in B.

   $$AB = \begin{bmatrix} 2\times 2 + 8\times 3 & 2\times 0 + 8\times 8 \\ 3\times 2 + 0\times 3 & 3\times 0 + 0\times 8 \\ 5\times 2 + 1\times 3 & 5\times 0 + 1\times 8 \end{bmatrix} = \begin{bmatrix} 28 & 64 \\ 6 & 0 \\ 13 & 8 \end{bmatrix}$$

   Not possible to calculate BA as B has two columns, but A has three rows.

   (b) BC and CB are both defined as both have two rows and two columns.

   $$BC = \begin{bmatrix} 14 & 4 \\ 69 & 30 \end{bmatrix} \neq CB = \begin{bmatrix} 20 & 16 \\ 21 & 24 \end{bmatrix}$$

4.

(a) $\begin{bmatrix} 0 & 2 \\ 36 & 20 \\ 16 & 3 \end{bmatrix}_{3\times2}$

(b) $\begin{bmatrix} 49 & 3 \\ 4 & 3 \end{bmatrix}_{2\times2}$

(c) $\begin{bmatrix} 3x+5y \\ 4x+2y-7z \end{bmatrix}_{2\times1}$

(d) $[7a+c \quad 2b+4c]$

Exercise 4.4

5. (e)

$$A = \begin{bmatrix} -2 \\ 4 \\ 7 \end{bmatrix}_{3\times1} \qquad B = \begin{bmatrix} 3 & 6 & -2 \end{bmatrix}_{1\times3}$$

$$C = AB = \begin{bmatrix} -6 & -12 & 4 \\ 12 & 24 & -8 \\ 21 & 42 & -14 \end{bmatrix}_{3\times3}$$

$$D = BA = [3\times-2+6\times4+-2\times7] = [4]_{1\times1}$$

7. In example 5, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}$. In which case,

$$\begin{aligned} x^TAx &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times1} \\ &= \begin{bmatrix} a_{11}x_1 & a_{22}x_2 \end{bmatrix}_{1\times2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times1} \\ &= \underbrace{a_{11}x_1^2 + a_{22}x_2^2}_{\text{Weighted sum of squares}} = \sum_{i=1}^{2} a_{ii}x_i^2 \end{aligned}$$

So x^TAx represents a weighted sum of squares where a_{11}, a_{22} are weights.

But now what if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. In this case,

$$
\begin{aligned}
x^T Ax &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \underset{2\times 2}{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}} \underset{2\times 1}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} \\
&= \begin{bmatrix} a_{11}x_1 + a_{21}x_2 & a_{12}x_1 + a_{22}x_2 \end{bmatrix} \underset{2\times 1}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} \\
&= a_{11}x_1^2 + a_{21}x_1x_2 + a_{12}x_1x_2 + a_{22}x_2^2 \\
&= a_{11}x_1^2 + (a_{21} + a_{12})\, x_1x_2 + a_{22}x_2^2
\end{aligned}
$$

So $x^T Ax$ no longer represents a weighted sum of squares.

You can check that the associative law i.e.

$$\left(x^T A\right) x = x^T (Ax)$$

will apply in both cases (after all, its a law!) as all products are possible.

Exercise 4.5

1.

$$A = \underset{2\times 3}{\begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}} \qquad b = \underset{3\times 1}{\begin{bmatrix} 9 \\ 6 \\ 0 \end{bmatrix}} \qquad x = \underset{2\times 1}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}$$

(a) $AI = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} \underset{3\times 3}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}} = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} = A$

(b) $IA = \underset{2\times 2}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}$

(c) $Ix = \underset{2\times 2}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(d) $x^T I = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$

4. Let's start with a 2×2 diagonal matrix

$$\begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11}^2 & 0 \\ 0 & a_{22}^2 \end{bmatrix}$$

$x = x^2$ for only $x = 0, 1$ so a_{11} and a_{22} can either be 0 or 1 . So we can have the following 2×2 idempotent diagonal matrices:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

More generally, for $n \times n$ matrix, there can be 2^n such matrices. This is because there will be n elements, each of which can take two values.

Exercise 4.6

2. $A = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -8 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}$

(a) $A + B = \begin{bmatrix} 3 & -4 \\ -1 & 4 \end{bmatrix}$

$A^T + B^T = \begin{bmatrix} 0 & -1 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ -8 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -4 & 4 \end{bmatrix}$

So, $(A + B)^T = A^T + B^T$

(b) $AC = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix}_{2\times 2} \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}_{2\times 3} = \begin{bmatrix} 24 & 4 & 4 \\ 17 & 3 & -6 \end{bmatrix}_{2\times 3}$ $C^T A^T = \begin{bmatrix} 1 & 6 \\ 0 & 1 \\ 9 & 1 \end{bmatrix}_{3\times 2} \begin{bmatrix} 0 & -1 \\ 4 & 3 \end{bmatrix}_{2\times 2} =$

$\begin{bmatrix} 24 & 17 \\ 4 & 3 \\ 4 & -6 \end{bmatrix}_{3\times 2}$ So, $(AC)^T = C^T A^T$.

6. $A = I - X\left(X^TX\right)^{-1}X^T$

   (a.) Say the dimension of X is $m \times n$. Then the dimension of $X^T_{n\times m}X_{m\times n}$ is $n \times n$. So the dimension of $\left(X^TX\right)^{-1}$ is also $n \times n$. This implies that the dimension of $X_{m\times n}\left(X^TX\right)^{-1}_{n\times n}X^T_{n\times m}$ is $m \times m$. Hence, X^TX and A must be square matrices, but X need not be square.

   (b.) To prove a matrix is idempotent, we need to show $AA = A$.

$$\begin{aligned}
AA &= (I - X\left(X^TX\right)^{-1}X^T)(I - X\left(X^TX\right)^{-1}X^T) \\
&= I - X\left(X^TX\right)^{-1}X^T - X\left(X^TX\right)^{-1}X^T + X\underbrace{\left(X^TX\right)^{-1}X^TX}_{I}\left(X^TX\right)^{-1}X^T \\
&= I - X\left(X^TX\right)^{-1}X^T - X\left(X^TX\right)^{-1}X^T + X\left(X^TX\right)^{-1}X^T \\
&= I - X\left(X^TX\right)^{-1}X^T = A
\end{aligned}$$

Exercise 5.1

3. (a) Yes
   (b) Yes
   (c) Yes
   (d) No; second row = $-2\times$ first row.

4. Yes, we get the same answer.

Exercise 5.2

1. (c)

$$\begin{aligned}
&8\begin{vmatrix} 0 & 1 \\ 0 & 3 \end{vmatrix} - 1\begin{vmatrix} 4 & 1 \\ 6 & 3 \end{vmatrix} + 3\begin{vmatrix} 4 & 0 \\ 6 & 0 \end{vmatrix} \\
=&8(0-0) - 1(12-6) + 3(0-0) \\
=&0 - 6 + 0 \\
=&-6
\end{aligned}$$

(e)

$$\begin{aligned}
&a\begin{vmatrix} c & a \\ a & b \end{vmatrix} - b\begin{vmatrix} b & a \\ c & b \end{vmatrix} + c\begin{vmatrix} b & c \\ c & a \end{vmatrix} \\
=&a\left(cb - a^2\right) - b\left(b^2 - ac\right) + c\left(ab - c^2\right) \\
=&abc - a^3 - b^3 + abc + abc - c^3 \\
=&-a^3 - b^3 - c^3 + 3abc
\end{aligned}$$

(f)

$$\begin{aligned}
&x\begin{vmatrix} y & 2 \\ -1 & 8 \end{vmatrix} - 5\begin{vmatrix} 3 & 2 \\ 9 & 8 \end{vmatrix} + 0\begin{vmatrix} 3 & y \\ 9 & -1 \end{vmatrix} \\
=&x(8y + 2) - 5(24 - 18) + 0(-3 - 9y) \\
=&8xy + 2x - 30 + 0 \\
=&8xy + 2x - 30
\end{aligned}$$

2. $|C_{13}| : 1 + 3 = 4$ is even so +
   $|C_{23}| : 2 + 3 = 5$ is odd so −
   $|C_{33}| : 3 + 3 = 6$ is even so +
   $|C_{41}| : 4 + 1 = 5$ is odd so −
   $|C_{34}| : 3 + 4 = 7$ is odd so −

3. Minor of a:

$$|M_{11}| = \begin{vmatrix} e & f \\ h & i \end{vmatrix} = ei - fh$$

   Cofactor of a:

$$|C_{11}| = (-1)^{1+1}\,|M_{11}| = |M_{11}|$$

   Minor of b:

$$|M_{12}| = \begin{vmatrix} d & f \\ g & i \end{vmatrix} = di - fg$$

   Cofactor of b:

$$|C_{12}| = (-1)^{3}\,|M_{12}| = -\,|M_{12}|$$

   Minor of f:

$$|M_{23}| = \begin{vmatrix} a & b \\ g & h \end{vmatrix} = ah - bg$$

Cofactor of f:

$$|C_{23}| = (-1)^5 |M_{23}| = -|M_{23}|$$

6. Minors of third row:

$$|M_{31}| = \begin{vmatrix} 11 & 4 \\ 2 & 7 \end{vmatrix} = 69, \quad |M_{32}| = \begin{vmatrix} 9 & 4 \\ 3 & 7 \end{vmatrix} = 51, \quad |M_{33}| = \begin{vmatrix} 9 & 11 \\ 3 & 2 \end{vmatrix} = -15$$

Cofactors: $|C_{31}| = |M_{31}|, |C_{32}| = -|M_{32}|, |C_{33}| = |M_{33}|$.

Exercise 5.3

1.

$$\begin{vmatrix} 4 & 0 & -1 \\ 2 & 1 & -7 \\ 3 & 3 & 9 \end{vmatrix} = 4 \begin{vmatrix} 1 & -7 \\ 3 & 9 \end{vmatrix} - 0 \begin{vmatrix} 2 & -7 \\ 3 & 9 \end{vmatrix} - 1 \begin{vmatrix} 2 & 1 \\ 3 & 3 \end{vmatrix}$$
$$= 4(9 + 21) - 0(18 + 21) - 1(6 - 3)$$
$$= 120 - 0 - 3 = 117$$

Interchanging rows and columns:

$$\begin{vmatrix} 4 & 2 & 3 \\ 0 & 1 & 3 \\ -1 & -7 & 9 \end{vmatrix} = 4 \begin{vmatrix} 1 & 3 \\ -7 & 9 \end{vmatrix} - 2 \begin{vmatrix} 0 & 3 \\ -1 & 9 \end{vmatrix} + 3 \begin{vmatrix} 0 & 1 \\ -1 & -7 \end{vmatrix}$$
$$= 4(9 + 21) - 2(0 + 3) + 3(0 + 1)$$
$$= 120 - 6 + 3 = 117$$

Interchange row 1 and 2:

$$\begin{vmatrix} 2 & 1 & -7 \\ 4 & 0 & -1 \\ 3 & 3 & 9 \end{vmatrix} = 2 \begin{vmatrix} 0 & -1 \\ 3 & 9 \end{vmatrix} - 1 \begin{vmatrix} 4 & -1 \\ 3 & 9 \end{vmatrix} - 7 \begin{vmatrix} 4 & 0 \\ 3 & 3 \end{vmatrix}$$
$$= 2(0 + 3) - 1(36 + 3) - 7(12 - 0)$$
$$= 6 - 39 - 84 = -117$$

You can verify the other two properties in the same way.

4. Multiplying every element of an nth-order determinant by k multiplies each of its n rows by k. Multiplying a single row by k multiplies the determinant by k, so doing it to all n rows gives $k^n|A|$.

5. (a)
$$\begin{aligned} &4\begin{vmatrix} 1 & -3 \\ 1 & 0 \end{vmatrix} + 1\begin{vmatrix} 19 & 1 \\ 7 & 1 \end{vmatrix} \\ &= 4(0+3) + 1(19-7) \\ &= \quad 12 + 12 = 24 \end{aligned}$$

Non-singular.

(b)
$$\begin{aligned} &5\begin{vmatrix} -2 & 1 \\ 0 & 3 \end{vmatrix} + 6\begin{vmatrix} 4 & 1 \\ 7 & 3 \end{vmatrix} \\ &=5(-6-0) + 6(12-7) \\ &= -30 + 30 = 0 \end{aligned}$$

Singular.

(c)
$$\begin{aligned} &7\begin{vmatrix} 1 & 4 \\ -3 & -4 \end{vmatrix} + 1\begin{vmatrix} 1 & 4 \\ 13 & -4 \end{vmatrix} \\ &=7(-4+12) + 1(-4-52) \\ &=56 - 56 = 0 \end{aligned}$$

Singular.

(d)
$$\begin{aligned} &-3\begin{vmatrix} 9 & 5 \\ 8 & 6 \end{vmatrix} - 1\begin{vmatrix} -4 & 9 \\ 10 & 8 \end{vmatrix} \\ &= -3(54-40) - 1(-32-90) \\ &= -42 + 122 = 80 \end{aligned}$$

Non-singular.

8. (a) False. While we can find the transpose of any matrix, the determinant is only defined for square matrices.

(b) False. Multiplying each element of an $n \times n$ by 2 will increase the determinant by 2^n times.

(c) A cannot be 0, but the determinant of A can. In any case, if $|A|$=0, A is singular.

Exercise 5.4

2. Note that

$$A^{-1} = \frac{1}{|A|} \operatorname{adj} A$$

Here, $\operatorname{adj} A = C'$ where $C = \left[\left|C_{ij}\right|\right]$.

(a)

$$|A| = 5 - 0 = 5$$

$$\operatorname{adj} A = \begin{bmatrix} |C_{11}| & |C_{21}| \\ |C_{12}| & |C_{22}| \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \operatorname{adj} A = \frac{1}{5} \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 1/5 & -2/5 \\ 0 & 1 \end{bmatrix}$$

(b)

$$|B| = -2 - 0 = -2$$

$$B^{-1} = \tfrac{-1}{2} \begin{bmatrix} 2 & 0 \\ -9 & -1 \end{bmatrix}$$

(c)

$$|C| = -3 - 21 = -24$$

$$C^{-1} = \tfrac{-1}{24} \begin{bmatrix} -1 & -7 \\ -3 & 3 \end{bmatrix}$$

(d)

$$|D| = 21 - 0 = 21$$

$$D^{-1} = \tfrac{1}{21} \begin{bmatrix} 3 & -6 \\ 0 & 7 \end{bmatrix}$$

3. (a) Step 1: Exchange the two diagonal elements.
Step 2: Multiply both the off-diagonal elements by -1.

(b) Step 3: Multiply the resulting matrix from steps 1 and 2 by $1/|A|$.

4. (a)

$$\begin{aligned}|E| &= 1\begin{vmatrix} 7 & 3 \\ 2 & 0 \end{vmatrix} + 1\begin{vmatrix} 4 & -2 \\ 7 & 3 \end{vmatrix} \\ &= 1(0-6) + 1(12+14) \\ &= -6 + 26 = 20\end{aligned}$$

$$\begin{aligned}\text{adj } E &= \begin{bmatrix} |C_{11}| & |C_{21}| & |C_{31}| \\ |C_{12}| & |C_{22}| & |C_{32}| \\ |C_{13}| & |C_{23}| & |C_{33}| \end{bmatrix} \\ &= \begin{bmatrix} 3 & 2 & -3 \\ -7 & 2 & 7 \\ -6 & -4 & 26 \end{bmatrix}\end{aligned}$$

$$E^{-1} = \frac{1}{|E|} adjE$$

(b)

$$F^{-1} = \frac{-1}{10}\begin{bmatrix} 0 & 2 & -3 \\ 10 & -6 & -1 \\ 0 & -4 & 1 \end{bmatrix}$$

(c)

$$|C_{11}| = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 \quad |C_{12}| = -\begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0 \quad |C_{13}| = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$|C_{21}| = -\begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} = 0 \quad |C_{22}| = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0 \quad |C_{23}| = -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = -1$$

$$|C_{31}| = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0 \quad |C_{32}| = -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = -1 \quad |C_{33}| = \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$G^{-1} = -1\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(d)

$$H^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

7. If a matrix is its own inverse, we would need that $A^2 = I$. This is true for the

identity matrix. There are other possibilities such as matrix G in exercise 4.

Exercise 5.4

6. (a)

$$A = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} \quad v = \begin{bmatrix} x \\ y \end{bmatrix} \quad b = \begin{bmatrix} 28 \\ 42 \end{bmatrix}$$

Since $Av = b \Rightarrow A^{-1}Av = A^{-1}b \Rightarrow v = A^{-1}b$

$|A| = 20 - 6 = 14$

$A^{-1} = \frac{1}{14}\begin{bmatrix} 5 & -3 \\ -2 & 4 \end{bmatrix}$

$$v = A^{-1}b = \frac{1}{14}\begin{bmatrix} 5 & -3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} 28 \\ 42 \end{bmatrix}$$
$$= \frac{1}{14}\begin{bmatrix} 140 - 126 \\ -56 + 168 \end{bmatrix} = \frac{1}{14}\begin{bmatrix} 14 \\ 112 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$

So $x = 1$ and $y = 8$.

(b)

$$A = \begin{bmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} 8 \\ 12 \\ 5 \end{bmatrix}$$

$$|A| = 4\begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} - 1\begin{vmatrix} -2 & 1 \\ 3 & 4 \end{vmatrix} - 5\begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix}$$
$$= 4(12 + 1) - 1(-8 - 3) - 5(2 - 9)$$
$$= 52 + 11 + 35 = 98$$

$$\text{adj}\, A = \begin{bmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{bmatrix}$$

$$x = A^{-1}b = \frac{1}{98}\begin{bmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{bmatrix}_{3\times3}\begin{bmatrix} 8 \\ 12 \\ 5 \end{bmatrix}_{3\times1} = \frac{1}{98}\begin{bmatrix} 196 \\ 490 \\ 98 \end{bmatrix}_{3\times1} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

Exercise 5.5

1 & 2. (a) $A = \begin{bmatrix} 3 & -2 \\ 2 & 1 \end{bmatrix} \rightarrow |A| = 7 \quad b = \begin{bmatrix} 6 \\ 11 \end{bmatrix}$

By Cramer's Rule:

$$A_1 = \begin{bmatrix} 6 & -2 \\ 11 & 1 \end{bmatrix} \rightarrow |A_1| = 28$$
$$A_2 = \begin{bmatrix} 3 & 6 \\ 2 & 11 \end{bmatrix} \rightarrow |A_2| = 21$$
$$x_1^* = \frac{|A_1|}{|A|} = \frac{28}{7} = 4$$
$$x_2^* = \frac{|A_2|}{|A|} = \frac{21}{7} = 3$$

Taking the inverse:

$$A^{-1} = \frac{1}{|A|}\,\text{adj} = \frac{1}{7}\begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}$$
$$\begin{bmatrix} x_1^* \\ x_2^* \end{bmatrix} = A^{-1}b = \frac{1}{7}\begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 6 \\ 11 \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 28 \\ 21 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

(b)

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -1 \end{bmatrix} \rightarrow |A| = -11 \quad b = \begin{bmatrix} -3 \\ 12 \end{bmatrix}$$

By Cramer's rule:

$$|A_1| = \begin{vmatrix} -3 & 3 \\ 12 & -1 \end{vmatrix} = -33 \rightarrow x_1^* = \frac{-33}{-11} = 3$$
$$|A_2| = \begin{vmatrix} -1 & -3 \\ 4 & 12 \end{vmatrix} = 0 \rightarrow x_2^* = \frac{0}{-11} = 0$$

By taking the inverse:

$$\begin{bmatrix} x_1^* \\ x_2 \end{bmatrix} = \frac{-1}{11}\begin{bmatrix} -1 & -3 \\ -4 & -1 \end{bmatrix}\begin{bmatrix} -3 \\ 12 \end{bmatrix} = \frac{-1}{11}\begin{bmatrix} -33 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

(c)

$$|A| = \begin{vmatrix} 8 & -7 \\ 1 & 1 \end{vmatrix} = 15$$

$$|A_1| = \begin{vmatrix} 9 & -7 \\ 3 & 1 \end{vmatrix} = 30 \quad \rightarrow x_1^* = \frac{30}{15} = 2$$

$$|A_2| = \begin{vmatrix} 8 & 9 \\ 1 & 3 \end{vmatrix} = 15 \rightarrow x_2^* = \frac{15}{15} = 1$$

Alternatively,

$$\begin{bmatrix} x_1^* \\ x_2^* \end{bmatrix} = \frac{1}{15}\begin{bmatrix} 1 & 7 \\ -1 & 8 \end{bmatrix}\begin{bmatrix} 9 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(d)

$$|A| = \begin{vmatrix} 5 & 9 \\ 7 & -3 \end{vmatrix} = -78$$

$$|A_1| = \begin{vmatrix} 14 & 9 \\ 4 & -3 \end{vmatrix} = -78 \rightarrow x_1^* = 1$$

$$|A_2| = \begin{vmatrix} 5 & 14 \\ 7 & 4 \end{vmatrix} = -78 \rightarrow x_2^* = 1$$

Alternatively,

$$\begin{bmatrix} x_1^2 \\ x_2^* \end{bmatrix} = \frac{-1}{78}\begin{bmatrix} -3 & -9 \\ -7 & 5 \end{bmatrix}\begin{bmatrix} 14 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

3. (a)

$$|A| = \begin{vmatrix} 8 & -1 & 0 \\ 0 & 2 & 5 \\ 2 & 0 & 3 \end{vmatrix}$$

$$= 8\begin{vmatrix} 2 & 5 \\ 0 & 3 \end{vmatrix} - 1\begin{vmatrix} 0 & 5 \\ 2 & 3 \end{vmatrix} = 48 - 10 = 38$$

$$|A_1| = \begin{vmatrix} 16 & -1 & 0 \\ 5 & 2 & 5 \\ 7 & 0 & 3 \end{vmatrix}$$

$$16\begin{vmatrix} 2 & 5 \\ 0 & 3 \end{vmatrix} - (-1)\begin{vmatrix} 5 & 5 \\ 7 & 3 \end{vmatrix} = 96 - 20 = 76$$

$$x_1^* = \frac{76}{38} = 2$$

$$|A_2| = \begin{vmatrix} 8 & 16 & 0 \\ 0 & 5 & 5 \\ 2 & 7 & 3 \end{vmatrix}$$

$$= 8\begin{vmatrix} 5 & 5 \\ 7 & 3 \end{vmatrix} - 16\begin{vmatrix} 0 & 5 \\ 2 & 3 \end{vmatrix}$$

$$= 8 \times -20 - 16 \times -10 = -160 + 160 = 0$$

$$x_2^* = 0$$

$$|A_3| = \begin{vmatrix} 8 & -1 & 16 \\ 0 & 2 & 5 \\ 2 & 0 & 7 \end{vmatrix}$$

$$= 8\begin{vmatrix} 2 & 5 \\ 0 & 7 \end{vmatrix} + 1\begin{vmatrix} 0 & 5 \\ 2 & 7 \end{vmatrix} + 16\begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix}$$

$$= 112 - 10 - 64 = 38 \rightarrow x_3^* = 1$$

(d)

$$|A| = \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= 1\begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}$$

$$= 0 + 2 + 2 = 4$$

$$|A_1| = \begin{vmatrix} a & 1 & 1 \\ b & -1 & 1 \\ c & 1 & -1 \end{vmatrix}$$

$$= a\begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix} - 1\begin{vmatrix} b & 1 \\ c & -1 \end{vmatrix} + 1\begin{vmatrix} b & -1 \\ c & 1 \end{vmatrix}$$

$$= 0 - (-b - c) + (b + c) = 2(b + c)$$

$$|A_2| = \begin{vmatrix} -1 & a & 1 \\ 1 & b & 1 \\ 1 & c & -1 \end{vmatrix}$$

$$= -1\begin{vmatrix} b & 1 \\ c & -1 \end{vmatrix} - a\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} 1 & b \\ 1 & c \end{vmatrix}$$

$$= -1(-b - c) - a(-1 - 1) + 1(c - b)$$

$$= b + c - 2a + c - b = 2(a + c)$$

$$|A_3| = \begin{vmatrix} -1 & 1 & a \\ 1 & -1 & b \\ 1 & 1 & c \end{vmatrix}$$

$$= -1\begin{vmatrix} -1 & b \\ 1 & c \end{vmatrix} - 1\begin{vmatrix} 1 & b \\ 1 & c \end{vmatrix} + a\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}$$

$$= -1(-c - b) - 1(c - b) + 2a$$

$$= c + b - c + b + 2a = 2(a + b)$$

$$x^* = \frac{b + c}{2} \quad y^* = \frac{a + c}{2} \quad z^* = \frac{a + b}{2}$$